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HMTraj: A Hybrid Mamba–Transformer Framework for Efficient Pedestrian Trajectory Prediction

1st Shiyi Tang
the Shien-Ming Wu School of Intelligent Engineering
South China University of Technology
Guangzhou, China
202320159824@mail.scut.edu.cn

2nd Kai Wu*
the Shien-Ming Wu School of Intelligent Engineering
South China University of Technology
Guangzhou, China
*whphwk@scut.edu.cn

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***Abstract*—Accurate and efficient pedestrian trajectory prediction is essential for safe motion planning in autonomous driving and mobile robotic systems. Although deep learning models have advanced this field, achieving a balance between prediction accuracy and computational efficiency remains a significant challenge, especially when jointly modeling long-range historical dependencies lying in pedestrian trajectories and complex social interactions among agents, hindering their practical applications. To address this, we propose HMTraj, a hybrid Mamba–Transformer framework for efficient pedestrian trajectory prediction. HMTraj employs a Motion-Mamba encoder operating on a compact set of representative motion patterns to efficiently capture diverse motion behaviors. Then, a Social-Attention decoder is used to selectively attend social interactions with relevant neighboring pedestrians. Experiments on the Stanford Drone Dataset (SDD) demonstrate that our method achieves competitive accuracy while maintaining superior computational efficiency with fewer parameters and lower inference time, validating its potential for providing reliable and practical trajectory prediction in resource-constrained, real-time applications.**

***Index Terms*—pedestrian trajectory prediction; hybrid neural network; mamba; social interaction modeling**

I. Introduction

The rapid advancement of autonomous driving technology fundamentally transforms the urban transportation system, enabling applications ranging from unmanned delivery robots [1] to self-driving vehicles [2]. Within the autonomous driving pipeline, the prediction module serves as a critical bridge between perception and planning, tasked with forecasting future states of surrounding agents to enable safe and proactive decision-making. Pedestrians, as the most vulnerable and unpredictable traffic participants, exhibit uncertain behaviors influenced by factors such as individual intent, social interactions, and environmental constraints. This introduces substantial uncertainty and potential risks, making accurate and reliable pedestrian trajectory prediction a key challenge for the overall success of intelligent transportation systems [3].

Early studies have designed physical models to represent pedestrian motion [4] and rule-based formulations to model social interaction [5]. While these hand-crafted methods are interpretable and lightweight, they are difficult to calibrate and generalize in crowded, complex, and unstructured environments. Deep learning-based models have significantly advanced the field by learning motion features and interaction behaviors directly from data. Recurrent Neural Networks (RNNs), particularly LSTMs [6], [7], have been widely used to extract spatio-temporal features from trajectories, but they often fail to capture long-range dependencies and suffer from inherent sequential processing inefficiencies. Transformer-based methods [8]–[10] further improve performance in capturing diverse behavioral features and complex social interactions by leveraging attention mechanism. However, their quadratic computational and memory complexity with respect to sequence length and the number of agents, hindering their applications under resource-constrained conditions. Since the movement of pedestrians is uncertain, pedestrians could take many socially acceptable trajectories in the future. Some works employ generative models or probabilistic models to generate a set of possible future trajectories for each pedestrian. The most typical methods are generative adversarial networks (GANs) [11] and conditional variational autoencoder (CVAE) [12].

Recently, a significant advancement in state space model, Mamba [13], has emerged as a promising alternative for efficient long-sequence modeling. Unlike conventional recurrent or Transformer architectures, Mamba introduces a selective mechanism that dynamically parameterizes the SSM based on the input, allowing the model to selectively propagate or discard information across long sequences in a content-aware manner. Combined with a hardware-efficient parallel algorithm, Mamba achieves linear computational complexity while maintaining strong performance on various tasks [14], [15]. These characteristics make Mamba particularly attractive for pedestrian trajectory prediction, where this task requires efficient reason over long spatio-temporal trajectory underlying diverse motion behaviors and complex social interactions under real-time constraints. However, directly applying Mamba to raw trajectory data presents challenges, as pedestrian trajectory data can be highly redundant and massive, and interactions among agents introduce complex social dependencies that are not trivially captured by purely temporal modeling.

To address these challenges, we propose HMTraj, a novel hybrid Mamba–Transformer framework for pedestrian trajectory prediction that combines a motion pattern generation preprocess with a Motion-Mamba encoder and a Social-Attention decoder, aiming to achieve a balance between

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prediction accuracy and computational efficiency. Firstly, we introduce a motion pattern generation module that constructs a compact set of representative motion patterns by clustering normalized pedestrians trajectories. This set covers common motion behaviors for all trajectories and structures the input space for subsequent modeling. Given an observed trajectory, we embed the motion pattern with the observation into a shared feature space to form the motion pattern embeddings, which are processed by a lightweight Motion-Mamba encoder to efficiently capture relationships among diverse motion patterns. Then, the Social-Attention decoder that employs masked cross-attention is used to selectively attend social interactions with relevant neighboring pedestrians, enabling flexible social interaction modeling. The main contributions of this work are summarized as follows:

- We propose HMTraj, a hybrid framework for pedestrian trajectory prediction that combines a Mamba-based encoder for efficient pedestrian motion modeling with a Transformer-based decoder for effective social interaction modeling. This lightweight model achieves a favorable trade-off between accuracy and efficiency.
- We introduce a motion pattern generation module that constructs a set of representative motion patterns by clustering normalized trajectories from data. This structured representation provides informative motion tokens for the motion encoder, effectively reduces redundancy in raw trajectory data and improves the prediction accuracy.
- Extensive experiments on the Stanford Drone Dataset (SDD) demonstrate that HMTraj achieves competitive prediction accuracy while requiring significantly fewer parameters and lower inference time than state-of-the-art methods, confirming its practical suitability for real-time autonomous systems.

II. Proposed Method

A. Problem Definition

The goal of pedestrian trajectory prediction task is to forecast future positions of pedestrians in a scene based on their historical observations. Formally, considering a scene contains N pedestrians, the observed historical trajectories of all pedestrians over T_{obs} time steps are denoted as: $X = \{(X_i)\}_{i=1}^{N}$, $X_i = \{(x_i^t, y_i^t)\}_{t=1}^{T_{\text{obs}}}$, where (x_i^t, y_i^t) represents the 2D coordinates of the ith pedestrian at time t. The prediction model is required to predict their future trajectories $\hat{Y} = \{\hat{Y}_i\}_{i=1}^{N}$, $\hat{Y}_i = \{(\hat{x}_i^t, \hat{y}_i^t)\}_{t=T_{\text{obs}}+1}^{T_{\text{obs}}+T_{\text{pred}}}$ for the following T_{pred} time steps. The set of prediction $\hat{Y}$ should closely approximate the ground truth trajectories $Y = \{(Y_i)\}_{i=1}^{N}$. For convenience, we denote the full trajectory length as $T = T_{\text{obs}} + T_{\text{pred}}$.

B. Model Overview

The overall framework of HMTraj is shown in Fig. 1. Our method initially generates a compact set of representative motion patterns by clustering all trajectories in the training dataset to structure the input sequence. For a given observed trajectory, we combine it with all patterns to embed into a shared feature space to create motion embeddings that present diverse and plausible future behaviors conditioned on the historical observation. This motion embeddings are then processed by a Motion-Mamba encoder, which operates selective scan along the pattern dimension using a lightweight Mamba block to capture long-range dependencies with linear complexity. Subsequently, these encoded motion features are fed into a Social-Attention decoder based on masked multi-head cross-attention to selectively aggregate information from socially relevant neighbors. Finally, the output of the decoder passes through a regression head to directly generate the future trajectory.

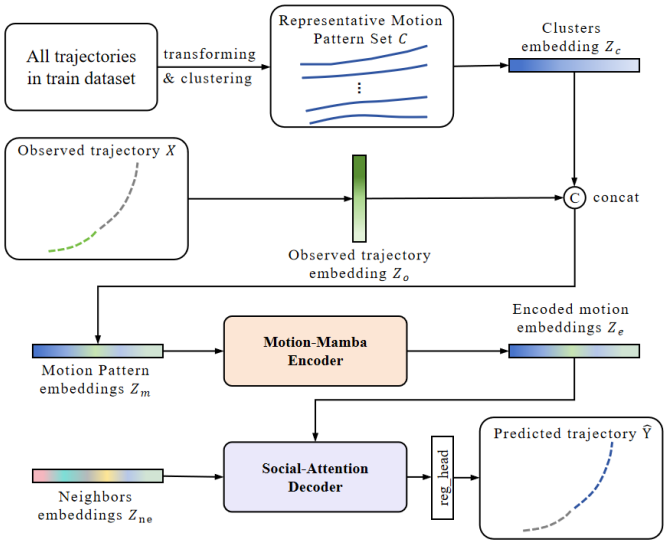


Fig. 1. Overall architecture of the proposed HMTraj

C. Motion Pattern Generation

Pedestrian motion over a short term typically conforms to a limited set of underlying patterns. To structure the input space and reduce the free space for subsequent prediction, we generate a compact set of representative motion patterns that covers common pedestrian behaviors observed in the training data. Specifically, each complete trajectory in the training set is split into an observed trajectory segment X_i with length T_{obs} and its corresponding future trajectory segment Y_i with length T_{pred}. The future segments Y_i are then normalized by translating their starting points to the origin and rotating them to align with the positive x-axis. This transformation removes absolute position and orientation variance, allowing the clustering process to focus on the relative shape of the motion. We then apply K-means algorithm to the normalized future segments to obtain L cluster centroids $C = \{c_1, c_2, \cdots, c_L\}$, where $C \in \mathbb{R}^{L \times T_{\text{pred}} \times 2}$ and c_i represents a prototypical future motion pattern.

For a given observation X, we project both the set of future motion patterns C and the observed trajectory X into a shared embedding space via two separate linear layers to obtain cluster embeddings $Z_c \in \mathbb{R}^{L \times D}$ and observed trajectory

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embedding $Z_o \in \mathbb{R}^D$ respectively. After broadcasting and concatenating the Z_c and Z_o along the pattern dimension, we construct the motion embedding input $Z_m \in \mathbb{R}^{B\times L\times D}$, which can be represented as:

$$Z_m = \text{concat}(Z_c, Z_o), \tag{1}$$

D. Motion-Mamba Encoder

The goal of the encoder is to model relationships and dependencies among these motion patterns in an efficient, content-aware manner. Following the Mamba1 architecture [13], we implement the Motion-Mamba encoder as a lightweight stack of Motion-Mamba blocks with a selective scan module operating along the pattern dimension. As shown in Fig. 2(a), the input feature is processed by a sequence of linear projection and convolution layer. Then, a selective SSM module with a residual connection is used to model long-range spatio-temporal information.

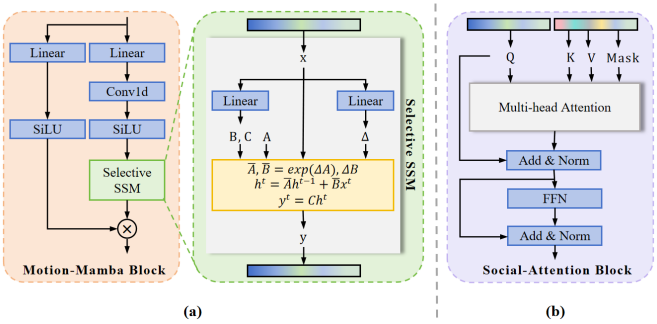


Fig. 2. (a) shows the Motion-Mamba block used in the encoder and (b) shows the Social-Attention block used in the decoder

Within the selective SSM module, which is the core component of Mamba, we first obtain three input-dependent matrices $B \in \mathbb{R}^{B\times L\times N}$, $C \in \mathbb{R}^{B\times L\times N}$, $\Delta \in \mathbb{R}^{B\times L\times D}$ derived from the input Z_m through linear layers, where N denotes hidden state size. Additionally, $A \in \mathbb{R}^{D\times N}$ is the system matrix. Specifically, the continuous parameters A, B are transformed into a discrete space using zero-order hold (ZOH) method with the timescale parameter Δ, formulated as follows:

$$\bar{A} = \exp(\Delta A) \tag{2}$$

$$\bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B \tag{3}$$

where $A, B \in \mathbb{R}^{B\times L\times D\times N}$. Then, we can compute the output by the SSM equation:

$$h^t = \bar{A}h^{t-1} + \bar{B}x^t \tag{4}$$

$$y^t = Ch^t \tag{5}$$

$$y = [y^1, y^2, ..., y^L] \tag{6}$$

where x^t represents the input at time step t, and y denotes the output of selective SSM.

By stacking a small number of such blocks, we obtain the encoded motion feature sequence $Z_e \in \mathbb{R}^{B\times L\times D}$, summarizing diverse pedestrian behaviors based on the current observation and then serving as the query input to the Social-Attention decoder, where it is further refined using neighbor information for interaction-aware trajectory prediction.

E. Social-Attention Decoder

The Social-Attention decoder models interactions between the target pedestrian and surrounding neighbors through a cross-attention mechanism. Suppose that the target pedestrian has N neighbors, whose observed trajectories over T_{obs} time steps are collected as X_{ne}, we embed these neighboring trajectories via a linear projection to obtain neighbors embeddings $Z_{ne} \in \mathbb{R}^{B\times N\times D}$. In the implementation, social interaction is modeled using a multi-head cross-attention layer as shown in Fig. 2(b). Inside the Social-Attention block, the query, key and value are derived from the encoded motion features and neighbors embeddings respectively, and reshaped to $Q \in \mathbb{R}^{B\times h\times L\times d_k}$ and $K, V \in \mathbb{R}^{B\times h\times N\times d_k}$, where h is the number of heads and $d_k = D/h$. Specifically, the critical mask matrix $M \in \{0,1\}^{B\times L\times N}$ is used to indicate which neighbors are valid (i.e., the neighbors within an interaction radius), ensuring the model only attends to valid interactions. This mask is expanded to $\mathbb{R}^{B\times h\times L\times N}$ and applied to the cross-attention scores calculation, which can be formulated as:

$$SocialAttn(Q, K, V, M) = softmax(\frac{QK^T}{\sqrt{d_k}} + M)V \tag{7}$$

The attended features are concatenated across heads, then passed through a residual connection and layer normalization, followed by a feed-forward network. By stacking several lightweight social-attention blocks, the Social-Attention decoder produces socially-aware motion features of size $\mathbb{R}^{B\times L\times D}$, which selectively integrate information from relevant neighbors and are subsequently fed into the regression head to directly generate future trajectory $\hat{Y}$.

F. Loss Function

We use the Huber loss between the predicted trajectory Y_i and the ground truth trajectory $\hat{Y}_i$ to train the overall framework, which is defined as:

$$\mathcal{L}_{tra} = \frac{1}{N}\sum_{i=1}^{N}\begin{cases}\frac{1}{2}(Y_i - \hat{Y}_i)^2 & \text{if } |Y_i - \hat{Y}_i| \le \delta \\ \delta(|Y_i - \hat{Y}_i| - \frac{1}{2}\delta) & \text{otherwise}\end{cases} \tag{8}$$

III. Experiments

A. Datasets and Implementation Details

We evaluate our proposed method on the Stanford Drone Dataset (SDD) [16], a large-scale benchmark for pedestrian trajectory prediction. SDD contains trajectories of 5,232 pedestrians across eight different scenarios captured by drone cameras from a bird's eye view. Each trajectory spans 3.2 seconds for observation (8 frames) and 4.8 seconds for prediction (12 frames).

We adopt two standard metrics for trajectory prediction performance. Average displacement error (ADE) measures the average ℓ_2 distance between the predicted trajectory and ground truth across all time steps. And final displacement

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TABLE I
PREDICTION ACCURACY COMPARISON ON SDD IN ADE/FDE. TEXT IN BOLD DENOTES THE BEST RESULTS.

Method	SocialLSTM [6]	SocialGAN [11]	SocialVAE [12]	AgentFormer [8]	MID [10]	HMTraj (Ours)
ADE	39.27	27.23	8.96	7.81	8.10	**7.72**
FDE	56.38	41.44	15.12	14.30	**11.72**	12.86

error (FDE) measures the ℓ_2 distance specifically at the final predicted position. Given the ground truth trajectory Y_i and predicted trajectory $\hat{Y}_i$ for i^{th} pedestrian, the metrics can be computed as:

$$\text{ADE} = \frac{1}{T_{pred}} \sum_{t=T_{obs}+1}^{T} \sqrt{(x_i^t - \hat{x}_i^t)^2 + (y_i^t - \hat{y}_i^t)^2}, \tag{9}$$

$$\text{FDE} = \sqrt{(x_i^T - \hat{x}_i^T)^2 + (y_i^T - \hat{y}_i^T)^2}, \tag{10}$$

Our implementation uses a single Motion-Mamba block and a single Social-Attention decoder block. The hyper-parameters in our model are set as $D = 128$, $E = 128$, $N = 64$, and $H = 4$. For motion pattern generation, we set the number of representative patterns to 100 for SDD. We employ the Adam optimizer with a learning rate of 0.001, batch size of 512, and train for 2000 epochs. All experiments are conducted using PyTorch on a single RTX 4090 GPU.

B. Analysis of Experimental Results

1) Accuracy: Table I demonstrates the quantitative comparison on SDD in terms of ADE/FDE. Classical RNN-based SocialLSTM [6] and GAN-based SocialGAN [11] show relatively large errors, reflecting their difficulty of capturing complex pedestrians behaviors with either purely recurrent modeling or adversarial training alone. SocialVAE [12] improves the performance by modeling the multi-modality of future trajectories, but its ADE/FDE are still noticeably higher than those of Transformer-based approaches (AgentFormer [8] and MID [10]). Our HMTraj obtains the best ADE of 7.72 and a competitive FDE of 12.86. Compared with AgentFormer [8], HMTraj reduces ADE from 7.81 to 7.72 and FDE from 14.30 to 12.86, indicating that our proposed method can capture both individual motion trends and social interactions more accurately. However, we notice that MID obtains the lowest FDE (11.72), but its ADE is worse than HMTraj (8.10 vs. 7.72), suggesting that HMTraj provides a better overall trade-off between average and final displacement errors while maintaining a much more compact and efficient architecture.

2) Efficiency: Fig. 3 compares the computational efficiency of different methods in terms of model size and inference time. Noticeably, Our achieves the fastest inference speed with only 0.0289s that significantly surpasses other methods, while remaining comparable lightweight model size to the previous smallest model SocialGAN [11] (46.8K vs. 46.3K). All these results validate the high efficiency and strong effectiveness of our proposed hybrid Mamba–Attention architecture for real-time, resource-constrained applications.

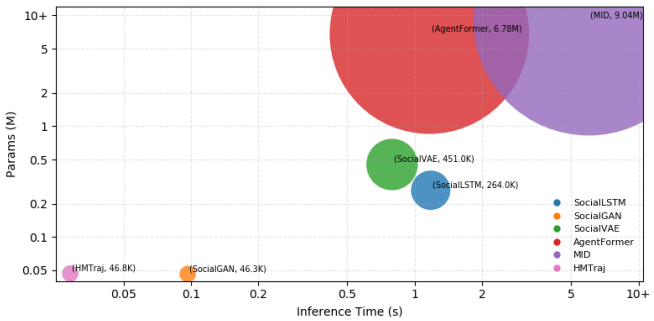


Fig. 3. Computational efficiency comparison on SDD.

C. Ablation Study

To assess the contribution of the main components in HMTraj, we design several ablation variants for the proposed framework. The results are shown in Table II, we replace the motion pattern generation (MPG) process by introducing L learnable pattern vectors that are randomly initialized, resulting in a significant 9.22%/21.41% increase in ADE/FDE. Moreover, we substitute the Motion-Mamba encoder (MME) and the Social-Mamba decoder (SMD) with a simple feed-forward network respectively, leading to a decrease in prediction accuracy. These results validate that each component is indispensable for improving the overall performance of the model.

TABLE II
ABLATION STUDY RESULTS OF DIFFERENT COMPONENTS ON SDD.

MPG	MME	SAD	ADE/FDE
×	✓	✓	16.94 / 34.27
✓	×	×	8.14 / 13.89
✓	×	✓	7.95 / 13.57
✓	✓	×	7.81 / 13.23
✓	✓	✓	**7.72 / 12.86**

IV. CONCLUSION

This paper proposes HMTraj, a hybrid Mamba–Transformer framework for efficient pedestrian trajectory prediction. By constructing a compact set of representative motion patterns, and modeling their relationships based on observation with a Motion-Mamba encoder, the proposed method provides a structured and lightweight way to capture diverse pedestrian motion behaviors. A Social-Attention decoder with masked cross-attention is used to selectively incorporate interactions with neighboring pedestrians to produce socially aware future

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trajectories. The experimental results on SDD dataset demonstrate that HMTraj achieves the best prediction accuracy in ADE of 7.72 over the baseline. At the same time, HMTraj maintains a very small model size with 46.8K parameters and the fastest inference speed with 0.0289 s, validating that the proposed method offers a favorable balance between accuracy and efficiency for practical pedestrian trajectory prediction.

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